Fiscal Year:	FY 2010	**Task Last Updated:**	FY 11/02/2009
PI Name:	Bingham, Gail E Ph.D.		
Project Title:	Validating Vegetable Production Unit Plants, Protocols, Procedures and Requirements Using Currently Existing Flight Resources		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Advanced human support technologies		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	(1) Developmental Biology		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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PI Web Page:			
City:	North Logan	**State:**	UT
Zip Code:	84341	**Congressional District:**	1
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2003 Space Biology 03-OBPR-01
Start Date:	01/01/2004	**End Date:**	09/29/2011
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	2	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA KSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: sometime in 2009, new grant was awarded with end date of 9/29/2011, per PI ; NNX09AV53G added. (Ed., 10/27/2011) NOTE: per PI, project transferred to KSC management (5/2008 report) NOTE: Per PI, end date is 7/31/2011; changed 8/24/2007 NOTE: corrected end date to 07/31/2007 per PI (1/07)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Garland, Jay (Life Support Group, Kennedy Space Center) Bates, Scott (Department of Psychology, Utah State University) Bugbee, Bruce (Department of Plants, Soils and Biometeorology, Utah State University)		
Grant/Contract No.:	NNX09AV53G ; NNJ04HG03G		

Performance Goal No.:	
Performance Goal Text:	
Task Description:	The primary objectives of the Lada VPU P3R experiments include advancing the Technology Readiness Level (TRL) of our plant species and support equipment, developing a Hazard Analysis and Critical Control Point (HACCP) plan for use of plant materials grown in the cabin, and measuring the non-nutritional value of various plants and plant presentation approaches. These studies will be carried out in both flight and ground studies. Plants Identify, flight-qualify and optimize the support requirements for the vegetable and flower plant varieties that are most likely to be utilized to meet flight crew needs (wants) and that fit within the hardware resource limitations that exist on the ISS and will exist on the CEV. This work will validate the technology readiness level of our available cultural practices to provide reliable, low-cost, stimulating products for crew well being. What plants can tolerate existing cabin ethylene levels without adding the cost of ethylene scrubbers? Will seed-producing plants be genetically stable over the 500 to 600 day mission? Food Safety Identify the threat levels and validate the procedures and protocols required to allow US astronauts to eat space-grown vegetables. Determine how to implement these procedures to maximize crew mental health benefits with minimum mission costs. What procedures and hardware need to be in place to assure crew health on a long flight? Non-Nutritional Value Quantify the mass value that should be assigned to the non-nutritional effects of plants in a spacecraft cabin. How much vegetable yield is required for a measurable boost to crew response under the isolation and stress of long-term space missions? How should this stress-relieving resource be presented (open or closed, single crew tended or multiple access) to maximize its value? Is this resource of value to all crew members, or is there a subset that prize the experience much more highly? The objectives will provide the data necessary to qualify plant products for authorized consumption on the ISS by US crewmembers. The Lada hardware and the cooperative agreement with IBMP have been shown to provide ideal, continuing opportunity to investigate many of the plant, procedure, protocol, and food safety requirement (P3R) issues associated with meeting crew livability requirements. The Lada VPU P3R uses existing hardware on the ISS (Lada) and leverages our cooperative agreement with the Institute of Biomedical Problems (IBMP) in Moscow.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Plants not only provide food, but for many they provide comfort and relaxation – a diversion from the stress of required activities. In the past, NASA has researched the option of including a Vegetable Production Unit (VPU) as a component of the life support system of their long-duration space flight vehicles. A previous road mapping exercise identified priority 1 and 2 Critical Path Risks for which plant materials and growth support activities could provide solutions. For at least some people, plants provide significant non-nutritional benefit to long-duration crews. These values are currently based only on anecdotal and untested observations that need verification. Part of the benefit may be a small fresh food source, which makes food safety issues important. The VPU study addresses these Critical Path risks, and in the process of monitoring microbial accumulations around Lada also contributes to crew health risk knowledge and reduction. Understanding the oxygen to water balance in the root zone applies not just in space, but in terrestrial agriculture as well. There is an acute need to optimize the efficiency of water use in agriculture, and leach-free experiments can be conducted only in space. This project will facilitate the understanding of how to better irrigate (precision watering) and fertilize different plants. The experiment also has considerable environmental implications as the ISS cabin is the ultimate setting for practicing recycling of plant, soil, water, and atmospheric resources.
Task Progress:	The accomplishments of the VPU research team during the 2009 year are as follows: • Three plant growth experiments completed on ISS during expeditions 17,19, and 20 with crops of barley, radish, and mizuna (two crops) • A third mizuna growth experiment to practice multiple harvest techniques started • Frozen samples of space grown barley, mizuna, radish (root and leaf) returned by shuttle from ISS and analyzed • Two spent root modules returned by shuttle from ISS and analyzed for microbial growth and root media conditions, A17 and A18 • Returned root module analyses and ground study data result in new recommendations for root module packing, recommendations included in packing of root module A21 for validation during upcoming ISS experiment • Successful collection of psychological data during Mars habitation chamber 105-day study in Moscow • Approvals obtained for on-orbit crew survey and post-flight crew survey of spaceflight participants who have worked with any plants in space • Successful ground radish grow-out completed to test sanitization techniques on microbial activity in growth chamber • Construction and delivery to Moscow for launch of three new Lada root modules A20, A21, and A22 The Lada VPU P3R experiment proposed schedule for downmass and frozen sample support as follows: • One root module and one frozen plant tissue sample scheduled for return Nov 2009 • Start of the first on-station, side-by-side, two-chamber plant growth experiment to evaluate root module packing improvements, Feb 2010 • Participation in Mars 502 study with plants in the simulation chamber starting in 2010

	• Two root modules and two frozen plant tissue samples, one refrigerated plant tissue sample scheduled for return Mar 2010
Bibliography Type:	Description: (Last Updated: 03/08/2018)
Articles in Peer-reviewed Journals	Gostimskii SA, Levinskikh MA, Sychev VN, Kokaeva ZG, Dribnokhodova OP, Khartina GA, Bingham G. "[The study of the genetic effects in generation of pea plants cultivated during the whole cycle of ontogenesis on the board of RS ISS]." Genetika. 2007 Aug;43(8):1050-7. Russian. PubMed PMID: 17958304 , Aug-2007
Articles in Peer-reviewed Journals	Gostimsky SA, Levinskikh MA, Sychev VN, Kokaeva ZG, Dribnokhodova OP, Khartina GA, Bingham G. "The study of the genetic effects in generation of pea plants cultivated during the whole cycle of ontogenesis on the board of RS ISS." Russian Journal of Genetics 2007 Aug;43(8):869-74. http://dx.doi.org/10.1134/S1022795407080066 , Aug-2007
Articles in Peer-reviewed Journals	Sychev VN, Levinskikh MA, Gostimsky SA, Bingham GE, Podolsky IG. "Spaceflight effects on consecutive generations of peas grown onboard the Russian segment of the International Space Station." Acta Astronautica 2007 Feb-Apr;60(4-7):426-32. http://dx.doi.org/10.1016/j.actaastro.2006.09.009 , Feb-2007
Articles in Peer-reviewed Journals	Marquit JM, Bates SC, Gushin VI, Sychev VN, Levinskikh MA, Podolsky IG, Marchant C, Bingham GE. "Testing crew responses to varied higher plant presentations in the Mars 500 mission simulation." Journal of Gravitational Physiology (in press). November 2009. , Nov-2009